

Understanding Team Adaptation: A Conceptual Analysis and Model

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This endeavor provides a multidisciplinary, multilevel, and multiphasic conceptualization of team adaptation with theoretical roots in the cognitive, human factors, and industrial–organizational psychology literature. Team adaptation and the emergent nature of adaptive team performance are defined from a multilevel, theoretical standpoint. An input–throughput–output model is advanced to illustrate a series of phases unfolding over time that constitute the core processes and emergent states underlying adaptive team performance and contributing to team adaptation. The cross-level mixed-determinants model highlights team adaptation in a nomological network of lawful relations. Testable propositions, practical implications, and directions for further research in this area are also advanced.

Keywords: teams, teamwork, team adaptation, adaptability, team effectiveness

It is not the strongest of the species that survives, nor the most intelligent, but rather the one most responsive to change.

—Charles Darwin

Change is an ever present reality of modern organizational living, and thus adaptation is essential. One mechanism by which organizations enhance their capacity to adapt is through the implementation of teams. Structuring work via teams rather than around individuals primes organizations to be more adaptive because collectives have a broader repertoire of capacities, experiences, and networks to draw on when engaging in performance change (Zaccaro & Bader, 2003). Adaptation lies at the heart of team effectiveness. The complex interdependence requirements of teams mandate that in order to be effective, members serve as compensatory systems for their fellow teammates, thus necessitating adaptation based on internal team cues. As the team collectively enacts behavioral strategies, individual differences in perceptions, tendencies, and assumptions within the team manifest as behavioral differences. These differences require members to dy-

namically adapt their actions within a prespecified area in order for individual performance and dyadic role exchanges to result in coordinative action.

Although teams can be well positioned to adapt (see Practical Implications section), little is known of this construct or the processes leading to team adaptation (Campbell & Kuncel, 2002). Therefore, this article provides a multidisciplinary, multilevel, and multiphasic conceptualization of team adaptation with theoretical roots in cognitive, human factors, and industrial–organizational psychology. Team adaptation and the emergent nature of adaptive team performance are defined from a multilevel, theoretical standpoint. An input–throughput–output model is advanced to illustrate a series of phases unfolding over time that constitute the core processes and emergent states underlying adaptive team performance and contributing to team adaptation (see Figure 1). Testable propositions, practical implications, and directions for further research are advanced.

Conceptualizing Team Adaptation

In an effort to illuminate the nature of team adaptation, we examined current organizational theory (e.g., general systems theory, open systems theory, sociotechnical systems theory) and existing definitions of adaptation at the team level and other related phenomena. During this process, definitions from multiple scientific domains were deliberately sampled (see Table 1). An effort was made to build on a number of perspectives so that the current endeavor could be more completely integrated with other conceptualizations of adaptation at the individual, team, and organizational levels, and empirically examined in a wider variety of research initiatives.

Building on this foundation, team adaptation is defined and centered in a multilevel nomological network of lawful relations. These steps, undertaken to illuminate the levels of the central constructs and linking processes, anchored the current effort to build a theory of team adaptation. Team adaptation is defined

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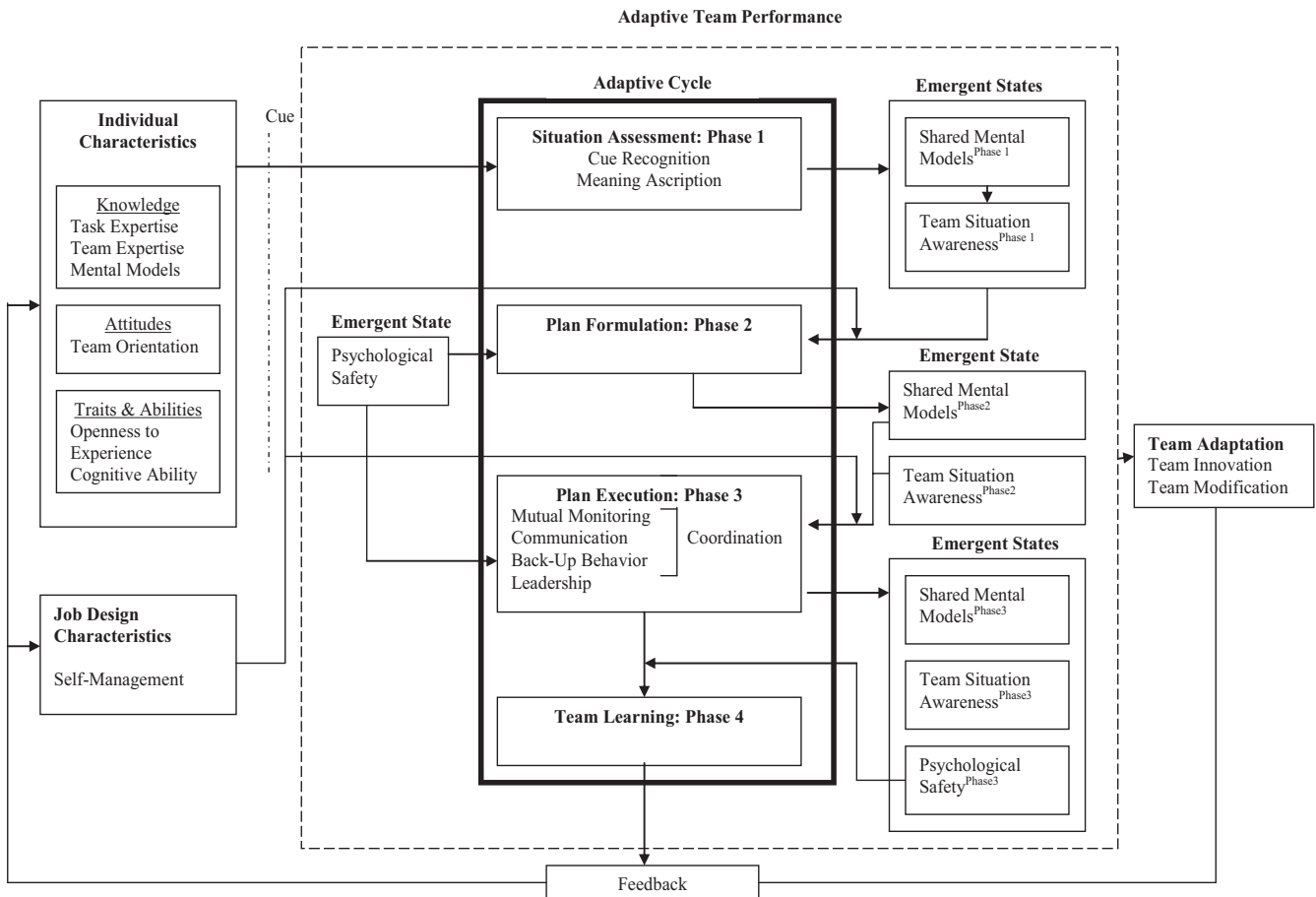


Figure 1. Input-throughput-output model of team adaptation.

herein as a change in team performance, in response to a salient cue or cue stream, that leads to a functional outcome for the entire team. Team adaptation is manifested in the innovation of new or modification of existing structures, capacities, and/or behavioral or cognitive goal-directed actions.

In order to understand team adaptation, it is fruitful to explore the relationships between team adaptation and other similar constructs. Indeed, the construct of team adaptation defined herein bears resemblance to extant team constructs such as team learning (see Edmondson, 1999), innovation (see De Dreu & West, 2001; West & Anderson, 1996), and problem management (see Tesluk & Mathieu, 1999). This subsection addresses how the current model builds on and extends each of these constructs by exploring key similarities and distinctions.

Building off prior work (e.g., Anderson, 1985), Edmondson (1999) argued that team learning can be viewed as the process by which relatively permanent changes occur in the behavioral potential of the group as a result of group interaction activities through which members acquire, share, and combine knowledge. It is the process by which team knowledge is gained through testing assumptions, discussing differences openly, forming new routines, and adjusting strategies in response to errors (Edmondson, 1999; Edmondson, Bohmer, & Pisano, 2001). Team learning also involves members jointly reflecting about their team processes and

behaviors (West, 2004). These activities enable team members to improve their collective understanding of a given situation and discover the consequences of previous actions, thereby helping them to detect changes in their operational environment (see Edmondson, 1999; Schippers, Den Hartog, Koopman, & Wienk, 2003). Engaging in these activities will result in knowledge being embedded within the team, which ultimately promotes performance improvement (Argote & Olivera, 1999).

On the surface, it might appear that team learning and team adaptation have considerable overlap. Upon deeper inspection, however, one sees that although related, these two constructs are disparate. Looking to Edmondson's and West's approach, team learning is primarily conceptualized as a cognitive event instilling knowledge that may, in turn, increase the behavioral repertoire of a team. This repertoire is, however, latent and may never manifest in functional change activity (LePine, 2003). It is the translation, differentiation, integration, and application of knowledge that allows teams to execute the cognitive and behavioral processes driving team adaptation (see Day & Lance, 2004). Members of adaptive teams utilize their pooled resources (i.e., knowledge gained from learning) to adjust their actions according to situational requirements. Thus, learning is an essential but insufficient condition for team adaptation. Although learning typically takes place *before* team adaptation, it can also be a consequence of

Table 1
Definitions of Adaptability and Adaptation

Date of origin	Authors	Definition
1995	Cannon-Bowers, Tannenbaum, Salas, & Volpe	The process by which a team is able to use information gathered from the task environment to adjust strategies through the use of compensatory behaviors and reallocation of intrateam resources
1999	Kozlowski, Gully, Nason, & Smith	Capability of the team to maintain coordinated interdependence and performance by selecting an appropriate network from its repertoire or by inventing a new configuration. Thus, adaptability refers to a metamorphic shift in the team network in the short term to deal with the performance demands of a nonroutine task.
2001	G. Klein & Pierce	Teams that are able to make the necessary modifications in order to meet new challenges
2001	Kozlowski, Toney, Mullins, Weissbein, Brown, & Bell	The generalization of trained knowledge and skills to new, more difficult, and more complex task situations
2003	Fleming, Wood, Dudley, Bader, & Zaccaro	Functional change in response to altered environmental contingencies and a higher order process that emerges from an integrated set of individual attributes
2003	LePine	Reactive and nonscripted adjustments to a team's system of member roles that contribute to team effectiveness
2004	<i>Merriam-Webster OnLine</i>	The act or process of adapting or the state of being adapted

adaptation, as in the case when teams learn competencies as a result of adapting.

Another construct closely related to team adaptation is team innovation. Team innovation has been defined in a number of ways (De Propris, 2002) but has often been described in terms of the number of subsystems involved, the locus of the innovation, different types of innovation (e.g., generation, architectural), or its characteristics (e.g., incremental, radical; Gatignon, Tushman, Smith, & Anderson, 2002). Team innovation has also been defined as (a) the creation and implementation of new ideas in the team setting for the purpose of improving the group and/or organization in some way (De Dreu & West, 2001; West & Farr, 1990) and (b) an improvement on existing products or processes due to use or experience (De Propris, 2002). Closer analysis of the construct can be seen in the work of Katila and Ahuja (2002), who argued that organizations search and solve problems in order to create new products (i.e., innovate), which in turn allows organizations to adapt. In iteratively progressing through this process, organizations can vary in the degree to which they exploit existing knowledge (i.e., search depth) as well as their exploration of new knowledge (i.e., search scope)—the former approach leading to incremental innovation and the later resulting in radical innovation.

In light of the advanced definition of team adaptation, the two constructs share key commonalities. Both are purpose driven, iterative, and the result of cognitive or behavioral actions carried out with the team's goals as the overarching priority (West & Anderson, 1996). In contrast to the advanced definition of team adaptation, team innovation is most often described as a process as opposed to an outcome (see Garcia & Calantone, 2002). A second distinction was made by Katila and Ahuja (2002), who argued that innovation is a precursor to adaptation. A third and more practical difference is that innovation may or may not lead to a functional outcome (i.e., realignment of performance), whereas the advanced definition of team adaptation requires this functionality. For example, advancements in information technology have allowed teams to become increasingly distributive. Innovative information technologies (e.g., video conferencing, Bluetooth) may indeed allow teams to overcome the performance decrements associated with distributed performance arrangements and thereby facilitate

team adaptation (a realignment of performance to prior acceptable levels). However, when misused or disregarded, the same innovative technologies may ultimately be an obstacle to team adaptation.

Finally, adaptive teams are often able to manage performance barriers effectively. Teams implement problem management techniques by either proactively preventing threats to performance or successfully removing barriers as they are encountered (Tesluk & Mathieu, 1999). For example, the constraints stemming from organizational policy, technology, or other aspects of a team's environment may occasionally pose hurdles to team effectiveness (see Goodman, 1986; Goodman, Ravlin, & Schminke, 1987). Nonetheless, adaptive teams have the capability to identify these performance hindrances and find ways to avoid or work around them.

This aspect of adaptive team performance parallels the existing concept of team problem management (see Tesluk & Mathieu, 1999). Problem management is similar, because it is in recognizing and managing obstacles that teams often demonstrate adaptation. However, teams do not always have to adapt in order to effectively manage a problem, as a solution can sometimes be generated without a concomitant modification or innovation by the team. Therefore, the current effort builds on and extends the notion of team problem management by analyzing the actual process by which teams change in circumstances mandating the need for adaptation.

In summary, adaptive responding is a general process underlying many team functions (e.g., dealing with performance hindrances, generating innovative solutions to problems, adopting new routines). In the next section the nomological network of team adaptation is fleshed out. Specifically, the multilevel, multiphasic, and cyclical nature of adaptive team performance and the core processes and emergent states that constitute it are described.

Adaptive Team Performance

Campbell (1990) stated that "performance is *not* the consequence(s) or result(s) of action; it is the action itself" (p. 704). Extending this argument to teams, Kozlowski and colleagues have argued that team performance has too often been conceptualized as the result of action, instead of the action itself (i.e., the longitudinal

enactment of processes) (Kozlowski & Bell, 2003). A similar differentiation between team performance and team outcomes is seen in the argument that the consequences of team performance include team member performance outcomes, team performance outcomes, individual changes, and team changes (Tannenbaum, Beard, & Salas, 1992).

In line with the above, team adaptation is conceptualized as the dependent variable of interest. The proximal temporal antecedents to team adaptation include a number of constructs, the core of which are labeled herein as adaptive team performance. Adaptive team performance is defined as an emergent phenomenon that compiles over time from the unfolding of a recursive cycle whereby one or more team members use their resources to functionally change current cognitive or behavioral goal-directed action or structures to meet expected or unexpected demands. It is a multilevel phenomenon that emanates as team members and teams recursively display behavioral processes and draw on and update emergent cognitive states to engage in change.

Similar to routine team performance, adaptive team performance can be conceptualized as either a global property of the team or a configural construct (Kozlowski, Gully, Nason, & Smith, 1999). The definition of adaptive team performance advanced above reflects an emergence process based on configural compilation. When conceptualized as a configural unit property, adaptive team performance can be seen as a continuously evolving phenomenon that compiles bottom-up across levels and time. Compilation is based on the assumptions of discontinuity or the configuration of different lower level properties to result in a higher level unit property (Kozlowski & Klein, 2000). Constructs that emerge through compilation do not represent shared properties across levels but rather are qualitatively different (i.e., constructs are characterized by patterns). It is the pattern of unique lower level team member and dyadic contributions that compile to characterize adaptive team performance (Kozlowski et al., 1999).

The specific manner in which team-level performance materializes and how that materialization is operationalized is contingent on organizational context, work-flow interdependencies, and other situational factors (K. J. Klein & Kozlowski, 2000). For example, members of a management team may each contribute relatively the same ideas during a brainstorming session to increase organizational productivity. This routine team task has a pooled level of interdependence whereby each member of the team engages in very similar behavior and makes an approximately equal contribution to the team outcome. In this example, team performance is essentially the sum of individual effort and can be operationalized in terms of mean performance. In contrast, adaptive team performance typically emerges as team members engage in different tasks and display different types and amounts of actions during performance. Therefore, the emergence of adaptive team performance is best captured by a patterned emergence model. For example, a management team consisting of division heads may have one member who provides a guiding vision for the team, whereas a second member provides backup behavior to a third team member busy gathering information about some recent contextual change. In this latter example, it is the pattern of individual-, dyadic-, and team-level contributions that make up team-level performance. These two examples illustrate team performance emerging along a continuum of discontinuity.

In sum, adaptive team performance emerges from a series of cognitive and behavioral actions carried out by team members. Team members draw from their individual and shared resources to detect, frame, and act on a cue or set of cues signaling the need for functional team-level change. As this occurs, several processes are enacted simultaneously, dynamically, and recursively. Enacted processes vary in type and amount, and as they are executed, individual and shared cognition and input factors are revised. This recursive process develops, progresses, and cycles over time, manifesting itself as adaptive team performance. In the next section, adaptive team performance is illustrated via a conceptual model of team adaptation.

A Conceptual Model of Team Adaptation

Several steps were taken to choose the constructs included in the cross-level mixed-determinants model of team adaptation displayed in Figure 1. First, team adaptation and adaptive team performance were defined, specifying the four core constructs characterizing the adaptive cycle: (1) situation assessment; (2) plan formulation; (3) plan execution, via adaptive interaction processes; and (4) team learning, as well as emergent cognitive states (i.e., shared mental models, team situational awareness, psychological safety), which serve as both proximal outcomes and inputs to this cycle. These processes and emergent states constitute adaptive team performance. The secondary variables (e.g., individual characteristics, job design characteristics) were chosen because they represent distal forces with respect to the adaptive cycle. Although these secondary constructs play a less central role in explaining adaptive team performance and team adaptation, they serve to illuminate a nomological net of lawful relations. To keep the framework as parsimonious as possible, only secondary variables that had supporting evidence linking them to team adaptation were included. In keeping with a multidisciplinary perspective, constructs were purposively sampled from a range of subdisciplines of psychology.

Theories of biological adaptation predominant in the natural sciences have been excluded, because they may have important consequences at specific levels of micromediation but not at the levels most pertinent to the present investigation. Constructs were also excluded at the multiteam system and organizational levels even though we remain convinced they contain phenomena pertinent to understanding team adaptation. The exclusion of phenomena at these levels was difficult, because teams are embedded in nested, intertwined systems of context and are thus affected by, and exert influence on, their environment. However, at this stage of theory building, a decision was made to focus solely on the bottom-up, emergent aspects of adaptive team performance that contribute to team adaptation rather than incorporating the simultaneous top-down process (see K. J. Klein & Kozlowski, 2000). Modeling the bottom-up process while ignoring the top-down process may seem arbitrary; however, this restricted focus was adopted because team adaptation is a relatively unexplored organizational phenomenon (see Campbell & Kuncel, 2002; K. J. Klein & Kozlowski, 2000). In fact, delineated insights will be used as the platform from which to launch future efforts to formulate a more comprehensive multilevel theory of team adaptation.

Figure 1 presents a testable cross-level mixed-determinants model of team adaptation. A cross-level mixed-determinants

model depicts the relationships between a single criterion at one level of analysis and its determinants, which originate at two or more levels of analysis (see Kozlowski & Klein, 2000; Yammarino & Dansereau, 2002). The model in Figure 1 illustrates the role of (a) cues, (b) individual characteristics, (c) job characteristics, (d) individual- and team-level adaptive interaction processes, and (e) emergent states in promoting team adaptation. These constructs are presented within an input-throughput-output model to show the proposed relationships among key variables and their relations to team adaptation. The model graphically depicts a theory of team adaptation, which at a broad level is composed of two categories of distal inputs (i.e., job design characteristics and individual characteristics) and two components of adaptive team performance (i.e., processes in the adaptive cycle and emergent cognitive states). The variables subsumed in these four categories directly and interactively determine team adaptation. Although the presented theory is argued to be most illustrative of teams that have moderate to high levels of task and/or outcome interdependence, it should generalize to teams with lower levels of interdependence. As interdependence decreases, the functionality of the components within the adaptive cycle would be expected to vary and be differentially weighted.

Adaptive Cycle and Emergent States

Figure 1 consists of twelve core variables that coalesce and compile across levels to constitute adaptive team performance. These core variables represent individual- and team-level process variables and the resulting emergent cognitive and attitudinal states. Emergent states are “constructs that characterize properties of the team that are typically dynamic in nature and vary as a function of team context, inputs, processes, and outcomes” (Marks, Mathieu, & Zaccaro, 2001, p. 357). Whereas process variables describe the nature of team member and team-level interaction, emergent states describe the cognitive, motivational, and affective state of the team (Marks et al., 2001). In the sections that follow, the nature of the four-phase adaptive cycle that is at the heart of adaptive team performance and ultimately team adaptation is addressed.

Phase 1: Situation Assessment—Emergent Cognitive States

Situation assessment describes “the human processes of gathering information (e.g., attention, pattern recognition, communication)” (Gutwin & Greenberg, 2004, p. 181). This individual-level cognitive process starts the adaptive cycle and consists of at least one team member scanning the environment in search of cues that could affect the success of the team’s mission. Individuals scan the environment to identify cues deemed relevant on the basis of prior experience and cognitive frameworks.

The identification of cues that signal a need for change is not a simple task. Habitual routines work against the recognition of such cues, and thus established entrainments must be brought to a conscious level in order to facilitate cue identification and meaning ascription. Whereas a failure on the part of the team is an obvious cue indicating some alterations may be necessary (Gersick & Hackman, 1990), cues can present themselves to a team in a variety of more subtle ways. For example, a cue may take the form

of a milestone such as the halfway point in the team’s project. There is evidence that teams engage in a regrouping process at the midpoint of their projects to increase their production in preparation for a deadline (Gersick, 1989). Waller, Zellmer-Bruhn, and Giambattista (2002) found this to hold regardless of the stability of the team’s deadline. Within these situations, the midpoint of the team’s project served as a trigger to adapt. Although teams differentially focused on temporal factors dependent on deadline type, actual transitions (i.e., adaptations) were most likely to occur at the midpoint.

The work of Louis and Sutton (1991) describes situations in which team members are more likely to identify cues that serve as triggers to adapt. Louis and Sutton identified the following conditions: (a) a situation that is experienced as unusual or novel; (b) a discrepancy, disruption, or unexpected failure; and (c) the occurrence of a deliberate initiative calling for an increased level of conscious attention to a cue (see also Gersick & Hackman, 1990; Okhuysen, 2001). For example, an intervention from the coaching staff of a sports team between plays (i.e., inside source), referees throwing flags (i.e., proximal outside source), or the league commissioner issuing fines (i.e., distal outside source) all can heighten awareness and provide the cues that indicate team adaptation is required.

Once a cue pattern is perceived, a series of recognition processes proceed, which result in the activation of a vast store of information later used in decision making (see G. Klein, 1993). This series of recognition processes draw on long-term memory to classify a situation or cue on the basis of existing mental models or schemata (Endsley & Smith, 1996). Meaning is assigned to cues and cue patterns by comparing them to existing knowledge structures. In the case of teams, the situation assessment process is likely to incorporate an additional step: the communication of that meaning to the rest of the team, circumstances permitting. This serves to transform the situation awareness and mental models that reside at an individual level to their corresponding team-level emergent states, shared mental models and team situation awareness. At an individual level, this temporarily “ends” the emphasis on the situation assessment process.

The manner in which shared mental models and team situation awareness emerge from their individual-level counterparts is dependent on situational characteristics, task interdependencies, existing norms, and one’s contextualized conceptualization of the latent construct. For example, shared mental models may emerge via either a composition or a compilation process, depending on how a researcher defines shared mental models. If shared mental models are defined as identical—in both amount and type—overlapping knowledge, then an isomorphic composition process of emergence is implied. In contrast, if shared mental models are defined as unique but compatible knowledge, then a configural compilation emergence process is implied (Kozlowski & Klein, 2000). Neither conceptualization is correct, per se, but rather each may be contextually appropriate. A similar argument can be made with respect to the emergence of team situational awareness. For example, given that only one member of the team is primarily responsible for conducting and transmitting the results of the situation assessment process to the rest of the team, the resulting team-level variable coalesced or emerged through composition. Note that this assumes that the rest of the team accepts the original version presented and that no real discussion follows, so that every

team member has an identical perspective of the current situation. Given the demonstration of restricted within-unit variance, aggregation to the team level can be represented by the mean or sum.

Conversely, if each team member is assessing the situation, there is the possibility that each will detect different cues and assign slightly different meanings dependent on existing knowledge structures and each member's vantage point. In this case, unique but compatible perspectives might compile into a congruent whole or team situation awareness. In this example, team situation awareness has emerged not through composition but via compilation. Configural properties can be represented by the minimum or maximum, variance, profile similarity, neural nets, and a number of other indices and techniques (see Kozlowski & Klein, 2000).

The situation assessment process and resulting team-level cognitive states promote adaptive team performance in at least two ways. First, prior to a team being able to make a functional adaptation there must be recognition that a cue pattern within the environment is indicating a need for change. By definition, the situation assessment process is the mechanism by which this cue may be recognized. Research suggests that the speed with which environmental changes are recognized and appropriate responses are enacted is related to subsequent team adaptability (Waller, 1999). Second, the emergent states that are the proximal outputs of the situation assessment process serve as the cognitive frameworks that allow team members to predict future system states with regard to member action as well as aspects of the current situation. Without team situation awareness, teams are not able to exhibit adaptive team performance owing to a lack of shared understanding with regard to the current situation. In the absence of shared mental models adaptive team performance is not possible, because members do not have compatible views of equipment, tasks, and team member roles and responsibilities, which allow members to adapt proactively.

Proposition 1: Situation assessment is positively related to the development of shared mental models within the first phase of the adaptive cycle.

Proposition 2: Situation assessment is positively related to the development of team situation awareness within the first phase of the adaptive cycle.

Phase 2: Emergent States—Plan Formulation

The second phase of the adaptive cycle is plan formulation. Planning has been argued to involve deciding on a course of action, setting goals, clarifying member roles and responsibilities within the context of a course of action, discussing relevant environmental characteristics and constraints, prioritizing tasks, clarifying performance expectations, and sharing information related to task requirements (Stout & Salas, 1993).

A number of variables serve as inputs to influence the quality of plan development (i.e., the degree to which, when executed properly, a plan reduces the gap between current and desired end state) and thereby team adaptation. Figure 1 pulls from the human factors (i.e., team situation awareness) and management literature (i.e., psychological safety) to identify two constructs that are most proximal to plan formulation.

Team situation awareness. Team situation awareness refers to a shared understanding of the current situation at a given point in time (Salas, Prince, Baker, & Shrestha, 1995). It is built on both the degree of shared understanding within a team, or shared mental models, and each individual member's situation awareness (based on preexisting knowledge bases and cue/pattern assessments) (Salas, Cannon-Bowers, Fiore, & Stout, 2001). Leveraging against individual situation awareness, team situation awareness can be argued to be composed of three separate levels that ascend in complexity: perception of environmental elements in relation to time and space (Level 1), understanding of which of these elements are noteworthy in relation to the team's goals (Level 2), and the ability to forecast future events in light of the current situation (Level 3) (Endsley, 1995).

During plan development, team members must maintain a shared perspective regarding which environmental elements are germane to the team's goals and, as such, should be given suitable attention. If members do not hold common vantage points, they may not attend to the right cues, or they may assign meaning that is incorrect within the context of the team's objectives. For example, some members may ignore a certain cue because they believe it is immaterial to the team's goals, whereas other members may take the cue very seriously. This could create inconsistency and frustration in addition to possibly impeding plan development, as members are not working from a common baseline with regard to environmental constraints.

Although situation awareness resides at both the individual and team levels, adequate team situation awareness does not necessitate that each member possess identical pieces of information. Rather, the members' situation awareness must sufficiently overlap to create a shared mental model regarding how to characterize and diagnose situations in the context of the team's objectives (Endsley, 1995). This information forms the baseline from which plan development originates. In addition, the extent to which a team is able to forecast future events in light of the current situation (i.e., possession of Level 3 team situation awareness) will allow the team to form plans that are proactive and flexible, instead of solely operating reactively.

Proposition 3: Team situation awareness is positively related to plan formulation during the second phase of the adaptive cycle.

Psychological safety. A second variable that will serve as an input to the team's plan development process is the degree to which psychological safety exists within the team. Team psychological safety has been defined as the shared belief that the team is safe for interpersonal risk taking (Edmondson, 1999). The concept, an extension of work conducted by Schein and Bennis (1965), reflects a team climate characterized by interpersonal trust and mutual respect. Edmondson (1999) argued that team psychological safety does not play a direct role in team performance but rather facilitates team members taking appropriate actions to accomplish work.

Team psychological safety's relationship to plan formulation originates in its role as an enabler of individual team members speaking up and offering contributions during plan development. This is particularly important given that one rationale for moving toward teams is that much organizational work has become too

cognitively complex for any one individual to successfully accomplish alone. Teams are effective in part because members often have different vantage points, as well as different levels and types of expertise and knowledge, that can be called on to address problems as they arise. Therefore, psychological safety contributes to quality plan development by promoting a climate whereby members feel free to question suggestions and decisions, in essence allowing members to play a type of devil's advocate. The promotion of such a climate may evolve through critical incidents that serve to set a precedent for team behavior. Mutual positive interaction among team members, especially in situations in which the team is under stress, may also serve to promote the mutual trust and respect necessary for psychological safety to emerge. Edmondson (2003) found that team leaders affect psychological safety through interpersonal activities, which serve to motivate the team and illustrate the importance of all members' inputs and downplay power differences.

Proposition 4: The level of psychological safety in a team is positively related to plan formulation during the second phase of the adaptive cycle.

The importance of plan formulation within the adaptive cycle lies within the fact that this process and the resulting product create a context in which commands and information requests made during plan execution take on meaning (Orasanu, 1990). This meaning is incorporated into existing knowledge structures (i.e., shared mental models) as the plan is communicated. Thus, plan formulation serves to revise shared cognition by serving as a proximal input into shared mental models.

Proposition 5: Plan formulation is positively related to the development of shared mental models within the second phase of the adaptive cycle.

Phase 3: Emergent States—Plan Execution

Plan execution involves an assortment of concomitant individual- and team-level processes that are enacted dynamically, simultaneously, and recursively (see Figure 1). For example, a football team may execute an offensive play in which the tight end cracks on a linebacker at the same time the wide receiver throws a key block that springs the running back for a long run to the end zone. This example is characterized by the enactment of both individual- and team-level processes as the offense engages in backup behavior and coordination during plan execution.

In contrast to the above example, both plan formulation and plan execution can also occur via the efforts of a single team member, especially when copious amounts of autonomy are the norm. For example, the quarterback may recognize the opposition suddenly shifting several of their linebackers to one side of the field, so that the defense can position for an all-out blitz as the play clock runs down. With no time to audible his intentions to the offense, the quarterback, having drawn on his expertise during the situation assessment process, relies on his physical quickness to sprint around the other end for a touchdown. Although this example does not involve dyadic role exchanges, the team still scored a touchdown. Another point raised by this example is that team adaptation results from a functional outcome that was achieved for the entire

team as opposed to just an individual team member, a fundamental distinction between individual adaptation and team adaptation.

In spite of the second example above, the plan execution phase typically involves a combination of individual- and team-level behaviors such as monitoring, backup, communication, leadership, and coordination in order to engage in adaptive team performance and achieve team adaptation. In the subsections that follow, the nature of these team member processes, and how they facilitate adaptive coordination during plan execution, is addressed. Following this, a treatment is given to how emergent cognitive states are both proximal inputs to and outcomes of plan execution.

Figure 1 identifies individual-level (i.e., mutual performance monitoring, backup behavior, communication, leadership) and team-level (i.e., coordination) processes that provide the basis for adaptive plan execution and thereby team adaptation. Generally speaking, carrying out a new plan requires communication and coordination of actions, whereas monitoring and backup behavior assist team members when cognitive or physical resources become depleted (e.g., in high-stress situations). Finally, team leadership is essential because the leader enacts processes that serve to structure member action, develop members, and assist members in maintaining and recreating the shared coherence needed to be adaptive (Kozlowski, 1998). Therefore, adaptive plan execution is typically successful only to the extent that teams and constituent members display the following processes: monitoring, backup behavior, communication, leadership, and coordination. The rationale for this argument is provided in further detail below.

Mutual performance monitoring. Mutual performance monitoring has been defined as team members' ability to "keep track of fellow team members' work while carrying out their own . . . to ensure that everything is running as expected and . . . to ensure that they are following procedures correctly" (McIntyre & Salas, 1995, p. 23). Mutual performance monitoring is primarily a cognitive action in which team members regularly observe the actions of their teammates and watch for mistakes, slips, lapses, errors, and performance discrepancies in an effort to catch and correct them in a timely manner.

Mutual performance monitoring contributes to a team's ability to adaptively execute a plan in several ways. First, it enables the recognition of when team members need assistance (Marks & Panzer, 2004). When team members need help, feedback in the form of verbal suggestions or corrective behaviors can assist in getting performance back on track by alerting team members to the adaptive action needed (Dickinson & McIntyre, 1997). Second, mutual performance monitoring contributes to a team's ability to execute a plan because it facilitates an awareness of the timing and pacing of team member action, which is needed for effective adaptive coordination (Kozlowski, 1998). Third, it has been argued that mutual performance monitoring facilitates higher levels of team situation awareness (Salas et al., 1995), which in turn contributes to plan execution because it provides the common ground needed to correctly adapt coordinated action. A team may also monitor its own progress toward goals and its interactions with the external environment and, in so doing, catch important cues signaling when to make adjustments in order to become realigned with overarching goals. If these cues slip by unnoticed, plan execution will falter and adaptation will be impossible.

Backup behavior. Porter et al. (2003) defined backup behavior as

the discretionary provision of resources and task-related effort to another member of one's team that is intended to help that team member obtain the goals as defined by his or her role when it is apparent that the team member is failing to reach those goals. (pp. 391–392)

For example, Member A may be monitoring Member B and realize that Member B is having some difficulty. Member A may either offer verbal instruction (i.e., backup behavior) or temporarily take over Member B's responsibilities (i.e., backup behavior) until the problem is resolved (Cannon-Bowers, Tannenbaum, Salas, & Volpe, 1995). It is the information gathered through mutual performance monitoring and expressed through feedback, backup behavior, and coordination that boosts the team from the sum of individual performance to the synergy of teamwork and in doing so promotes plan execution and team adaptation.

Although backup behavior can occur in response to specific requests for help or from the recognition that there is a workload distribution problem in the team, its role in promoting adaptive team performance may vary depending on the actual team need for the offered backup behavior. Porter et al. (2003) found that when underutilized individuals back up the individual whose capacity is being surpassed, teams can dynamically adjust and perform at a level that could not have been otherwise achieved by individuals acting alone. In contrast, in circumstances where there is not a legitimate need for backup behavior, the provision of such behavior can actually detract from adaptive team performance because it leads to redundant instead of complementary behavior. The provision of legitimate backup behavior promotes plan execution and thereby adaptive team performance by providing the feedback and coaching that allows members to adjust their actions in a timely manner. In essence, backup behavior contributes to the capability to adapt coordination processes, which is central to plan execution. In turn, flexibly executing developed plans and revising action when needed is vital to facilitating adaptive team performance in changing situations and environments.

Communication. Communication has been defined as "the process by which information is clearly and accurately exchanged between two or more team members in the prescribed manner with proper terminology; the ability to clarify or acknowledge the receipt of information" (Cannon-Bowers et al., 1995, p. 345). Communication plays a pivotal role in the adaptive cycle. First, communication is essential in the development and updating of the shared knowledge structures that serve to guide adaptive action. Second, communication provides the foundation for effective monitoring behavior; if a team member monitors fellow members' actions yet never speaks up in the form of feedback or backup behavior, the monitoring is not functional for the team.

Leadership. The benefits of leaders and leadership in teams have been well documented (see Burke et al., 2005). Team leaders are valuable because they serve as coordinators of operations, as liaisons to external teams or management, and as guides for setting the team's vision (Zaccaro & Marks, 1999). The work of several leadership researchers has suggested that team leaders can play a key role in facilitating a team's propensity to adapt by choosing how and when to intervene to promote review and revision of procedures and methods (e.g., Gersick & Hackman, 1990; Hackman & Wageman, 2005). Recent work by Hackman and Wageman (2005) has identified specific times in task performance when

coaching interventions by the leader are most likely to have intended effects. Of most relevance to the current context is the argument that consultative coaching, minimization of mindless adoption of routines in uncertain environments, and fostering of strategies well aligned with current conditions are most helpful when provided at the midpoint (i.e., during the execution phase), whereas other forms of leader intervention (e.g., motivational, educational coaching) are most helpful at the beginning and completion of performance, respectively.

Although team leaders can play a key role in a team's propensity to adapt, the process of team leadership contributes to the team's capability to adapt coordination processes and thereby facilitates flexible plan execution not by handing down solutions to team members but rather by facilitating team problem solving through cognitive processes, coordination processes, and the team's collective affective status (Salas, Burke, & Stagl, 2004). In this vein, the leadership processes enacted by the team leader can play a key role in promoting the conditions required for adaptive action during the execution phase of the adaptive cycle.

Taking a slightly different tack, there is a growing interest in the idea that teams may benefit from shared leadership (Brown & Giolia, 2002; Burke, Fiore, & Salas, 2003; Klenke, 1997; Pearce & Conger, 2003). Shared leadership has been defined as "the transference of the leadership function across team members in order to take advantage of member strengths (e.g., knowledge, skills, attitudes, perspectives, contacts, time available) as dictated by either environmental demands, or the developmental stage of the team" (Burke et al., 2002, p. 105). A team with this type of shared leadership system should be better able to adapt to internal and external changes. For example, if a team leader must discontinue his or her duties, the team may adaptively shift the leadership role and corresponding leadership processes to another member. As leadership functions are dynamically transferred among team members, effectiveness is heavily dependent on the smooth transference among team members (i.e., coordinated leadership transfer).

In order for the transference of leadership functions to occur effectively, there are a few prerequisites. First, the process is dependent on the possession of shared knowledge structures (i.e., shared mental models, team situation awareness). Shared mental models, particularly the team mental model, reduce the workload associated with shared leadership by assisting members in the determination of *where* the leadership function needs to transfer. The team mental model is especially helpful, as it contains information related to team member characteristics and the individual and collective requirements for effective team interaction. Situation assessment and the resulting team situation awareness (when shared) enable the effective transference by providing a context-sensitive assessment of the current situation and guiding the determination of *when* the leadership function should transfer. Second, the process is dependent on an attitudinal factor in that there needs to be a climate created such that members feel it is acceptable to have fluid leadership roles. This requires that members be comfortable with taking guidance and cues from different people, dependent on who is currently assuming the leadership function. The confluence of these factors is a smooth transference of leadership from one member to another and the facilitation of adaptive team behaviors.

Coordination. Whereas the processes described up to this point have occurred at an individual level, coordination is a team-level phenomenon. Coordination involves the team organizing the sequencing and timing of its actions (Marks et al., 2001). Given the interdependent nature of teams, ongoing coordination of actions is a requisite within any team environment. However, within environments that are fairly dynamic, not only must teams coordinate but they must do so in an adaptive fashion. In essence, the adaptive application of the behavioral processes that facilitate coordinated action (e.g., backup behavior, mutual performance monitoring, communication, leadership) is what allows effective teams to capitalize on potential synergies or process gains.

The manner in which the team member processes combine to create adaptive coordinated action is dependent on task, team, and situational characteristics. Specifically, we adopt a functional approach to posit that the weighting of the individual team member processes will be different depending on the aforementioned characteristics. For example, it is expected that within a highly interdependent team, the team member processes mentioned above would be weighted equally, whereas in a less interdependent team, backup behavior and/or mutual performance monitoring may be less central to success.

Research suggests that teams can be adaptive in the form of coordination they enact. Specifically, team coordination may take on one of two forms—implicit or explicit—depending on the complexity and stress inherent in the task (Entin & Serfaty, 1999; Serfaty, Entin, & Volpe, 1993). Implicit coordination requires that members draw from their shared mental models to anticipate and meet the needs of their teammates without being asked. Implicit coordination is particularly useful during periods of high stress, because it serves to decrease workload (Entin & Serfaty, 1999). However, during routine tasks, explicit or overt coordination may be more appropriate.

Proposition 6: Mutual performance monitoring, backup behavior, leadership, and communication will be positively related to coordinated action during the third phase of the adaptive cycle.

In addition to the processes constituting plan execution, there are also proximal inputs that affect plan execution (see Figure 1). Two such inputs are shared mental models and psychological safety. Shared mental models arise out of the communication of the plan and serve to guide member action. Possession of shared mental models of the equipment, task, and team members is essential to the enactment of team processes such as mutual performance monitoring and backup behavior. Specifically, to effectively engage in mutual performance monitoring and backup behavior, members must have an understanding of their team members' jobs and be both willing and able to provide and seek assistance when needed. Without an accurate shared understanding of each other's roles, members cannot effectively catch mistakes or lapses, nor can they provide feedback essential to backup behavior.

Proposition 7: Shared mental models are positively related to plan execution within the third phase of the adaptive cycle.

Although shared mental models are important, the level of team situation awareness at the beginning of plan execution is also an

important factor in how the plan will be executed. As team situation awareness involves an overlapping perception of various environmental elements and an understanding of which of these elements are important in relation to the team's goals, team situation awareness will influence the cues that initially elicit team member attention as the plan is being executed, as well as how members adapt their internal coordination mechanisms.

Proposition 8: Team situation awareness is positively related to plan execution within the third phase of the adaptive cycle.

Although shared mental models and team situation awareness are indeed important, the level of psychological safety is also related to how effectively a plan can be executed. Within Phase 2 of the adaptive cycle, psychological safety was described as a shared belief reflecting interpersonal trust, mutual respect, and a comfortableness with interpersonal risk taking. As such, existing levels of psychological safety within the team will affect plan execution via the degree to which mutual performance monitoring is accepted. Similarly, it is predicted to play a role in the degree to which backup behavior is offered and accepted. Finally, the level of psychological safety existing within the team may affect a team's comfortableness with implicit coordination. As implicit coordination relies on shared knowledge structures it also represents a degree of risk, as team members are assuming that the knowledge structures that are shared are also accurate.

Proposition 9: Psychological safety in a team is positively related to plan execution within the third phase of the adaptive cycle.

As depicted in Figure 1, the interaction processes within the third phase of the adaptive cycle will serve to update existing affective and cognitive states (i.e., shared mental models, team situation awareness, psychological safety). When a team applies a plan of action, both the team and its surroundings are affected. These changes are then integrated within individual cognitive structures, serving as an updating mechanism. When the environmental and team changes that occur as a result of plan execution are shared, either through communication or joint perception, it culminates in the shared knowledge structures indicative of shared mental models and team situation awareness. These new cognitive structures incorporate any new cues and their potential effects on the team.

Proposition 10: Plan execution is positively related to the formation of shared mental models within the third phase of the adaptive cycle.

Proposition 11: Plan execution is positively related to the formation of team situation awareness within the third phase of the adaptive cycle.

In addition to altering a team's reservoir of shared cognition, team member interaction during plan execution will also serve to influence existing levels of psychological safety. If interaction proceeds in a positive manner in terms of the factors influencing psychological safety (i.e., trust, mutual respect, and comfortableness with interpersonal risk), then the level of psychological safety will remain the same or potentially increase. However, if a nega-

tive team interaction ensues (i.e., a member gets chastised), then the level of psychological safety within the team may decrease. With this caveat in mind, as psychological safety is indicative of a shared climate or set of perceptions, there is a high probability that preexisting levels are fairly resilient, and thus it may take a major event or several small events over time to detract from current levels.

Proposition 12: Plan execution is positively related to the formation of psychological safety within the third phase of the adaptive cycle.

Phase 4: Team Learning

The final phase in the adaptive cycle is team learning. Levitt and March (1988) conceptualized learning as the outcome of a process that consists of encoding inferences from history into the routines that guide behavior. Building on this, Edmondson (1999) conceptualized learning at the group level as “an ongoing process of reflection and action, characterized by asking questions, seeking feedback, experimenting, reflecting on results, and discussing errors or unexpected outcomes of actions” (p. 354). Team learning facilitates the development of knowledge and contributes to the ability of members to improve their collective understanding of a given situation. The team discovers the consequences of previous actions, the mechanisms by which unintended consequences can be prevented, and the manner in which courses of action can be revised. This knowledge is then used by members to scan their environment for changes (see Edmondson, 1999; Schippers et al., 2003).

Given that team learning has been argued to be a process made up of team members jointly and openly discussing errors and unexpected outcomes in order to subsequently revise cognition and behavior accordingly, we predict that the degree of psychological safety that exists within the team will be a key moderator of the extent to which team learning occurs following team interaction (i.e., plan execution). For maximal learning to occur, members must be willing to expose themselves interpersonally by openly discussing mistakes, mismatches, and alternative viewpoints. As psychological safety has been defined as the degree to which members perceive the environment to be safe for interpersonal risk taking (Edmondson, 1999), we predict that when the psychological safety within the team is low, there is less of a chance that the team will actually learn on the basis of their previous interactions as opposed to situations in which there exists a high level of psychological safety present within the team.

Proposition 13: Psychological safety will moderate the relationship between plan execution and team learning, the final phase of the adaptive cycle.

Job Design Characteristics

The advanced team adaptation model also highlights the importance of job design characteristics. Specifically, the model illustrates the moderating role of team self-management. The concept of team self-management is closely linked with several overlapping constructs, including autonomy (Hackman & Oldham, 1980), behavioral discretion (Cannon-Bowers, Salas, & Blickensderfer,

1998), and empowerment (Mohrman, Cohen, & Mohrman, 1995). In fact, team self-management is the group-level analogue of team member autonomy (Campion, Medsker, & Higgs, 1993). Common to these constructs is the idea that members and teams have a certain degree of freedom to determine how and when to coordinate their inputs.

Implicit in Figure 1 is the conviction that “pushing the power down” will, given the presence of enabling competence, allow teams to more appropriately tweak their activities to the particular contingencies encountered. Empowering and enabling members and teams with the leeway to make near real-time decisions will enhance adaptation via the effectiveness of plans, in terms of both quality and timeliness, which can be formulated and executed. Moreover, self-management also flows from adaptation. In fact, the operational latitude afforded to team members and teams by key organizational stakeholders can increase in scope over time. For example, a team that makes a series of successful innovations may be given self-management over scheduling.

When team members are given little autonomy, they fail to experience a sense of responsibility for their performance and thus are less likely to engage in critical processes (Hackman & Oldham, 1980). In fact, there is evidence suggesting a positive relationship between team self-management and team effectiveness (Campion et al., 1993; Campion, Papper, & Medsker, 1996). A possible explanation suggests that when a team controls its own functions, it has the opportunity to alter its processes and strategies when necessary. Moreover, control at an individual level creates a sense of responsibility for work outcomes, and thus team members are more likely to want to engage in these behaviors (Hackman & Oldham, 1980).

The advanced model illustrates that self-management, or the amount of decision latitude teams are empowered with, will moderate the relationship (a) between team situation awareness and plan formulation and (b) between shared mental models and plan execution. For instance, a team may have an accurate team situation awareness that indicates the need for change but have limited freedom to actually make a plan to remedy the situation. Similarly, a team may formulate a plan but may be constrained in what they can actually implement. Without additional autonomy, the team would likely fall back on half measures and thus be less effective in completing the adaptive cycle. Hence, self-management not only enables teams to make critical decisions as to the best methods for goal accomplishment but allows for the discretion to follow through on those decisions. Thus, as the level of self-management within a team increases, the team will be more likely to successfully formulate and carry out effective plans and thereby ultimately be more adaptive.

Proposition 14: Self-management moderates the relationship between team situation awareness and plan formulation.

Proposition 15: Self-management moderates the relationship between shared mental models and plan execution.

Individual Characteristics

Teams are composed of individuals, and therefore it makes intuitive sense that team members' attributes will contribute to team member performance and thereby to team performance (see

Figure 1). Specifically, adaptive teams will be composed of individuals with high levels of task expertise, team expertise, team orientation, openness to experience, and cognitive ability. Furthermore, teams with members who hold accurate, flexible mental models will have enhanced levels of team adaptation.

The effects of these individual differences are often moderated and sometimes mediated by a number of processes (e.g., situation assessment) and emergent states (e.g., situation awareness). Furthermore, the importance of member characteristics to any given performance episode will likely be contingent on a host of context- and situation-specific factors such as the technology used by a team, the level of interdependencies, and task design characteristics (see Ackerman & Humphreys, 1990). Although these complex contingencies warrant investigation, space constraints limit the current discussion to why each of these six team member characteristics indirectly promotes team adaptation by directly enhancing team members' adaptive capacity (i.e., adaptability) to engage in the process of situation assessment.

Task Expertise

In the advanced model of team adaptation, task knowledge (i.e., expertise) is proposed to be a determinant of situation assessment. However, in order for task expertise to contribute to situation assessment it cannot consist of mere organized, static knowledge (Kozlowski, 1998) or a tunnel vision perspective that results in team members performing their own duties in a perfunctory manner while disregarding the larger framework within which the team is operating. Rather, members must be exceptionally familiar with not only the task but also the principles that underlie various aspects of the task (LePine, 2003). Simply stated, members should know what to do, how to do it, and why it should be done. It is also crucial for each team member to possess an awareness of both the situation and how his or her role meshes with the teammates' roles within that context. Deep comprehension of task principles and members' roles ensures that the team will have a broad repertoire of responses at its disposal. As a result, the team will not have to rely on one or two methods for solving a particular problem but will draw from members' combined expertise to choose among many courses of action.

The first phase of the adaptive cycle includes the two steps of noticing and assigning meaning to an environmental cue. These two steps may occur at the individual level; then, through the medium of team processes, a plan is formulated and implemented. For instance, a member may notice a problem with the team's performance. That member brings it to the attention of the rest of the team—thereby shifting the team's understanding of the current situation—and together the members generate a solution. Thus, it is essential that each member is skilled at recognizing pertinent cues signaling the need for adaptation. This recognition entails, to a great degree, the dynamic application of a person's expertise with both the task and the team of which he or she is a part. In addition, knowledge of both of these domains allows the team to maintain an accurate, shared understanding of events unfolding, events that are either internal or external to the team itself.

Proposition 16: Task expertise is positively related to situation assessment.

Team Expertise

In addition to task expertise, a strong case can also be made that those individuals who are highly knowledgeable about their fellow teammates will pick up on internal cues better than those who are unfamiliar with their teammates. For instance, an individual who is fairly accustomed to working with his or her teammates may easily recognize cues suggesting that another member is not behaving or performing as usual. These cues suggest adaptive behaviors may be necessary, such as providing task-related or personal assistance to the faltering member or compensating for his or her performance in some way. Team members who have worked together for a relatively long period of time and under varying circumstances may be more likely than members of a newly formed team to notice slight alterations in the ways their colleagues are functioning together. Thus, members who share a history may retain a more accurate awareness of the team's internal status and thereby more quickly recognize pertinent cues and ascribe a deeper, richer meaning to those cues detected.

Proposition 17: Team expertise is positively related to situation assessment.

Mental Models

Mental models are dynamic, simplified, cognitive representations of reality that team members use to describe, explain, and predict events. Moreover, team members draw on mental models to guide their interactions with others and with the elements that make up their systems of operation. Team members also use mental models to interpret and integrate new information (Rouse & Morris, 1986). In fact, it has even been asserted that mental models are the basic structure of cognition (Johnson-Laird, 1983).

At least five types of mental models have been identified, including equipment, task, team, team interaction, and problem/situation (Cannon-Bowers, Salas, & Converse, 1993). Each model influences what team members perceive in their operating environment and how recognized cues are reacted to. It is beyond the scope of the present initiative to specify the exact nature of how each of these five mental models affects team member situation assessment. However, members with accurate and flexible mental models will be more effective at identifying pertinent cues and more precise in the meaning they assign to those cues during the ongoing process of situation assessment.

Proposition 18: Mental models are positively related to situation assessment.

Team Orientation

It is also apparent that despite their importance, member expertise and other such knowledge structures are insufficient for team adaptation. Each team member may have a great deal of expertise; however, if those team members do not cooperate by subordinating their own proximal goals and sharing their knowledge and skills at the appropriate times in order to achieve the collective's goals, mistakes and ultimately failure will ensue. Adaptive teams are composed of team players who work well with others, seek others' input, contribute to the team's outcome, and enjoy team membership (Goodwin, O'Shea, Driskell, Salas, & Ardison, 2004). Indi-

viduals who possess these characteristics and who have a general preference to work in teams when given the opportunity are said to have a "team orientation" (Goodwin et al., 2004).

Substantiating the above noted line of thinking, team performance has been linked to the frequency with which members engage in collective, cooperative behaviors, including accepting and receiving input and suggestions from teammates (Driskell & Salas, 1992). Moreover, when collectively oriented team members disagreed, they attended to the problem and sought a solution, whereas noncollective team members largely discounted others' input and ignored the resulting group discord (Driskell & Salas, 1992). Research also suggests that teams composed of individuals who prefer group work experience higher levels of member satisfaction (Shaw, Duffy, & Stark, 2000) and exhibit more cooperative behaviors (Eby & Dobbins, 1997; Wagner, 1995). In fact, evidence suggests that team cooperation mediates the relationship between team orientation and team performance (Eby & Dobbins, 1997).

Although the relationships between team orientation and the processes contributing to team adaptation have not to date been specifically examined, a proclivity for group work will be positively related to the processes that make up the adaptive cycle. Specifically, members with a preference to accept input from others are more likely to consider a wide range of perspectives about internal and external team cues and thereby assign similar meaning to contextual events. Essentially, team members with a preference for teamwork are likely to perceive and interpret stimuli in a similar fashion.

Proposition 19: Team orientation is positively related to situation assessment.

Openness to Experience

The advanced model emphasizes the importance of team member openness to experience in promoting member situation assessment. Openness to experience is an interpretative name given to one of five factors that are repeatedly found from structural analyses of common descriptive words. The five domains, or spheres of function, are often labeled Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness to Experience. Each domain is itself multidimensional, as each of the five domains is composed of "multifaceted collections of specific cognitive, affective and behavioral tendencies" (Costa & McCrae, 1985, p. 23).

Although many of the characteristics mentioned above could be central to facilitating situation assessment, the specific focus herein is on openness to experience. Openness to experience is characterized by a range of descriptors including imaginative, curious, open-minded, and original (Costa & McCrae, 1992). Clearly, these characteristics would contribute to noticing and assigning meaning to cues that are often subtle and fleeting. Furthermore, open individuals engage in self-monitoring, which is essential for learning in novel environments (Blickle, 1996). Thus, member openness to experience serves to facilitate the situation assessment process.

Accumulating theoretical and empirical research indirectly supports the above assertion (see Barrick, Stewart, Neubert, & Mount, 1998; Barry & Stewart, 1997; Driskell, Hogan, & Salas, 1987; Driskell & Salas, 1992; Neuman, Wagner, & Christiansen, 1999; Porter et al., 2003). In fact, research suggests that openness to

experience is positively related to team member decision-making performance and more strongly so after a contextual change forces an adaptive response (LePine, Colquitt, & Erez, 2000). LePine and colleagues' (2000) results can be interpreted to suggest that open individuals are less likely to become entrenched in routines and are more accepting of novel solutions to problems.

Proposition 20: Openness to experience is positively related to situation assessment.

Cognitive Ability

Guion (1998) defined cognitive abilities as the "abilities to perceive, process, evaluate, compare, create, understand, manipulate, or generally think about information and ideas" (p. 124). Although many conceptualizations of cognitive factors exist (see Cattell, 1963; Sternberg, 1985), most scholars acknowledge both a general factor, *g*, and specific factors. In fact, research results suggest that scores on tests of specific abilities can explain incremental variance in organizational criteria beyond that explained by *g* alone (Thorndike, 1986).

Cognitive ability is critical to facilitating adaptive team performance and team adaptation via the processes members are capable of engaging in and the extent to which they are successful in those activities. Intuitively, it would seem that there would be a direct relationship between several specific abilities and the core processes proposed within the adaptive cycle. For example, Guion's (1998) conceptualization of creativity as an ability to see opportunities in unplanned events would seem relevant to team member situational assessment. In fact, it has been suggested that holding all other factors constant, teams composed of high-ability individuals will outperform teams made up of individuals with low abilities (Tannenbaum et al., 1992).

Evidence supporting these assertions is voluminous. For example, research suggests that team member cognitive ability is related to member adaptive decision-making performance (LePine et al., 2000). These findings also suggest that member cognitive ability is more strongly related to performance after an unforeseen change takes place, adding to a growing body of evidence suggesting that cognitive ability is more important for performance in novel than routine environments (Hunter & Hunter, 1984). Adopting a higher level of analysis, LePine (2003) found that teams composed of high cognitive ability members were more adaptive than teams composed of members with low cognitive ability. It seems teams composed of high cognitive ability members can better adjust their role structure to conform to an unexpected change in task context and this role modification enhances decision accuracy.

Although the above evidence is intriguing, the nature of the relationships between *g*, cue recognition, and meaning ascription have yet to be explored. Therefore, in spite of this preliminary empirical evidence, it is essential that the specific links between *g* and the adaptive activities (e.g., noticing cues, ascribing meaning) constituting situation assessment be examined further. Moreover, the relationship between *g* and situation assessment should be investigated in contexts that mandate both proactive and reactive adaptation to routine and novel demands.

Proposition 21: General mental ability is positively related to situation assessment.

Measuring Team Adaptation

Adaptation is change, and although this point is both simple and sublime, measuring and analyzing the change inherent to team adaptation in the workplace is anything but. Complexities notwithstanding, any thorough investigation of team adaptation would be deficient without some attention directed at the measurement and analysis of multilevel longitudinal change. However, given the scope of conceptualizing, measuring, and analyzing change over time, and particularly simultaneously modeling change at both the individual and team levels, the current discussion is restricted to a few of the most pertinent issues as they apply to the advanced input-throughput-output model of team adaptation. The reader is directed to several recent comprehensive reviews for additional information (see Chan, 1998; Day & Lance, 2004; K. J. Klein et al., 2000).

In a prior section of this article, team adaptation was defined as a change in team performance, manifested in the innovation of new or modification of existing structures, capacities, and/or behavioral and cognitive goal-directed actions, in response to a salient cue or cue stream. Implicit in this supposition is the notion that an adaptation results in a functional outcome for the entire team, as opposed to just an individual team member. But what is meant by *functional*, and how does one know when a change manifested in terms of innovation or modification is *functional* for a given team? To help answer these questions we turned to the *Merriam-Webster OnLine* (2004) dictionary to define this term and found that something is functional when it is used to contribute to the maintenance or development of a larger whole. Following this line of thinking, an adaptation, as carried out via either innovation or modification, is functional if it ultimately contributes to the maintenance or development of the larger whole (e.g., a given team).

In clarifying what it means to be functional, we have also purposely described the two primary aspects that require scrutiny in order to assess team adaptation: (a) the maintenance of the team and (b) the growth of the team. The maintenance of team performance is advanced as the primary challenge encountered when a team detects a cue pattern indicative of a misalignment between the team's current performance level and the demands imposed by its operational context. Secondary indicators of team adaptation include the development of team member task expertise at the individual level and shared emergent cognitive states at the team level. A discussion follows of some of the methodological and analytical issues relevant to measuring primary and secondary indicators of team adaptation.

Several recent research initiatives have conceptualized adaptation in a general fashion as a longitudinal process whereby a mismatch or misalignment between output and demands is minimized (Chan, 2000; Chen & Ployhart, 2004; Kozlowski et al., 1999; LePine et al., 2000). These efforts complement the current endeavor undertaken to more fully illuminate the nature of this "longitudinal process," because collectively, they imply a straightforward approach to measuring team adaptation. Specifically, adaptation is functional to the extent that it minimizes the decrements in task performance resulting from environmental changes and thereby contributes to the maintenance of the larger system.

As noted by Chen and Ployhart (2004), the performance trajectories of adaptive individuals typically follow a nonlinear pattern where some level of acceptable performance is followed by a

transition period denoted by a decline in performance due to misalignment, which is in turn followed by a subsequent realignment (i.e., adaptation) that serves to increase performance. This unfolding pattern of an initial acceptable level of performance followed by an unacceptable level of performance due to misalignment and subsequently followed by an adaptation and return to an acceptable level of performance can be illustrated by plotting team performance levels as a function of time. Once plotted, team performance should follow a negatively accelerated monotonic curve (Chen & Ployhart, 2004). The inflection of the curve (i.e., rate of change) is indicative of team adaptation, in that teams with steeper curves are more adaptive in responding to a cue stream signaling the need for change. The later portion of this performance trajectory is similar in form to a learning curve, whereas the first portion indicates baseline acceptable performance with a sudden negative deceleration.

Conceptualizing team adaptation in this fashion presents a number of methodological and analytical issues that must be addressed in order to fruitfully empirically investigate team adaptation. Although adaptation is the outcome of a longitudinal process, it has often been studied via the use of cross-sectional research designs in conjunction with commonly used descriptive and inferential statistical techniques. However, this traditional approach is inadequate to accurately model the complexities of individual, team, and organizational adaptation. Fortunately, statistical approaches are increasingly available (e.g., mixed effects modeling [MEM], latent growth modeling [LGM], multiple indicator latent growth modeling [MLGM]), which can be coupled with longitudinal research designs to appropriately model the change inherent to team adaptation (see Bliese & Ployhart, 2002; Chan, 1998; Day & Lance, 2004). A longitudinal approach coupled with MEM, LGM, or MLGM can be used to simultaneously model team member and team adaptation, illuminating performance change at multiple levels in an organization.

In addition to modeling changes in the level of team performance due to adaptation, MEM, LGM, and MLGM can also be applied to modeling secondary changes that occur concomitantly as teams seek homeostasis with a dynamic environment. Specifically, these statistical techniques can be used to analyze changes in emergent cognitive states or individual characteristics, such as those resulting from team member adaptation. For example, MLGM grounded in confirmatory factor analysis can be used to model longitudinal change of team member task expertise as a result of engaging in adaptive team performance. Such an approach is an effective means for controlling for random measurement error, tracking qualitative and quantitative change, and assessing individual differences in change.

Practical Implications

This article opened with a quote from Charles Darwin that spoke to the importance of adaptation for the very survival of all species. Although nothing so grandiose in scope is claimed for the implications of the model advanced herein, fostering team adaptation remains important to the effective functioning and thereby viability of organizations. In fact, the implications of the advanced model of team adaptation cross a wide spectrum of organizational functioning to extend to system design, information technology design, job design, assessment for selection, socialization efforts,

individual development, team development, and, more broadly, the facilitation of change at multiple levels in the conceptual space.

Though each of these areas of human capital management is noteworthy in its own right, space constraints preclude a detailed treatment of each domain, and thus the focus of the current endeavor is restricted to a discussion of the implications of the advanced framework for team and team member development. This focus has been chosen because individuals, teams, and organizations are increasingly immersed in a fluid and global context that mandates the need for continuous employee development (see Day, Zaccaro, & Halpin, 2004; Rutkowski, Steelman, & Griffith, 2004). Therefore, in this section examples are offered of organizationally sponsored training interventions that can be used to target a sample of key constructs argued herein to constitute adaptive team performance and team adaptation.

Team Member Training and Adaptive Team Performance

At the individual level, learning to solve novel problems is essential to developing adaptive expertise (Smith, Ford, & Kozlowski, 1997). To accomplish this, trainees should be required to generalize their skills to new and more fluid situations. Unfortunately, much of the extant literature focuses on training for simple, routine tasks while largely ignoring complex, dynamic tasks (Kozlowski et al., 2001). This is regrettable, because repetitious, routine practice will likely be inadequate at best for training adaptive skills. Following this line of thinking, DeCroock, Paas, and Van Merriënboer (1998) reported that individuals trained on a complex task with barriers impeding them from obtaining quick proficiency demonstrated better transfer of training than those trained without obstacles. This could mean that individuals trained with no barriers to mastering the task fall into a repetitious pattern for task completion and that this routine actually hampers performance when any type of novel facet is subsequently introduced.

In concurrence with this logic, evidence suggests that trainees who are allowed to make errors while learning a task achieve a better grasp of task principles than trainees who are prevented from committing errors (Frese et al., 1991). Errors signify deviations from the goal and serve to catch the learner's attention because they represent unanticipated events (Smith et al., 1997). Introducing errors gives trainees practice in handling new, unexpected events, keeping them in an active learning mode while they gain a handle on fundamental task principles. Thus, it is theorized that giving trainees the opportunity to make errors will assist them in reaching a higher level of understanding of the task than if they are prevented from making mistakes.

Another method for enhancing adaptive team performance and ultimately team adaptation is to foster cognitive flexibility in team members by presenting task principles from many angles and points of view, and especially from fellow teammates' perspectives. The objective is to impart flexible knowledge representations that can be extended and molded by members to fit new situations (see Spiro, Feltoich, Jacobson, & Coulson, 1992). Scenario-based training (SBT) is one instructional strategy that can be used to illuminate multiple perspectives on the same issue while promoting active learning. This form of training uses work simulations as the basic training tool. Methods that promote team member active learning (e.g., advance organizers, analogies, guided discovery, error-based training, metacognitive instruction, learner control,

and sequenced mastery goals) can also be embedded within SBT frameworks to help foster adaptive team performance (Kozlowski, 1998).

A slight variation of traditional SBT implementation may also be effective for developing team member mental models and thereby team adaptation. SBT can be coupled with embedded situational judgment tests in order to provide trainees the opportunity to reflect on one event at a time followed by an open forum discussion of that event's problem characteristics, contextual constraints, and proven solutions (Fritzsche, Stagl, Salas, & Burke, 2006). This low-fidelity approach to SBT could be a cost-effective means of facilitating the development of flexible mental models, because each team member is privy to other team members' thinking about the nature of, and solution to, the encountered problem. Similarly, Marks and colleagues suggest the use of team interaction training to impart orientation and mapping functions for improving the quality of team member mental models and adaptive performance (Marks, Zaccaro, & Mathieu, 2000).

One final approach to facilitating team adaptation by targeting team members for development is via the use of instructional strategies that promote skill in situation assessment. Several instructional strategies exist that can be used to develop team member skill in situation assessment, and accumulating empirical evidence supports these approaches. For example, contrasting cases training can be used to help team members recognize when cues signal the need to adaptively engage in teamwork processes during judgment tasks (Fritzsche et al., 2004). Team adaptation and coordination training (TACT) can be used to promote adaptation under stressful circumstances (Entin, Serfaty, & Deckert, 1994). Several components of TACT consist of instruction in recognizing signs and symptoms of stressful situations and training team members to provide situation updates. A third training approach for enhancing team member situation assessment, team situation assessment training for adaptive coordination, emphasizes linking cues, cue pattern recognition, and adaptive team behaviors (Martin-Milham & Fiore, 2004). A fourth training intervention that can be used to help members be more effective at engaging in situation assessment was advanced as one component of a 1-week classroom-based course (White, Dorsey, Pulakos, & Mueller-Hanson, 2003). White and colleagues presented three training modules that target mental adaptability, interpersonal adaptability, and leading an adaptable team.

Team Training and Adaptive Team Performance

Teams of experts do not necessarily evolve into expert teams, and thus targeted interventions are required to achieve this status. As these efforts pertain to facilitating adaptive team performance and team adaptation, teams have often attempted to overplan, anticipate all situations likely to be faced, define responses in as much detail as possible, and/or wait for encountered ambiguity to subside. Unfortunately, each of these approaches has limitations and can serve as a barrier to adaptation (Schmitt & Klein, 1996). Fortunately, a wide range of interventions can be called on to facilitate team adaptation, ranging from participation in communities of practice to exposure to operational stretch assignments. Expert teams can also develop from the guided discussion, modeling, and practice that occur in organizationally sponsored formal

training programs. The remainder of this subsection presents training strategies and guidelines for enhancing team adaptation.

In terms of team training, a number of guidelines have been advanced to promote team adaptation. For example, it has been proposed that adaptive skills should be taught first at the individual level, progress to role dyads, and then finally extended to the role networks that constitute the entire team (Kozlowski, 1998). In this way, team members are first learning to master their own skills while discovering how their roles mesh with their teammates' roles. In this regard, computer-based instructional systems can be designed with synthetic avatars to prompt trainees to engage in metacognitive strategies during training, thereby actively cuing trainees to think about their activities in relation to others.

Relevant literature also suggests that mastery goals rather than performance goals should be set for both members and the team as a whole early in training (Kozlowski, 1998). That is, for both individual and team training, mastering the task should be the first priority, rather than reaching a precisely prespecified level of performance. This point echoes prior empirical evidence suggesting that goal achievement in the early stages of declarative learning can be detrimental to skill acquisition for trainees (Kanfer & Ackerman, 1989).

In addition to the guidelines noted above, team adaptation can also be facilitated by training for both internal and external adaptations (G. Klein & Pierce, 2001). Teams should be trained to make modifications and innovations to their structure, systems, and procedures in order to develop the response repertoires and climate of trust that can facilitate dynamic internal adaptations. Similarly, blending whole-task with part-task training, developing communication workarounds, and imparting problem-solving routines will help teams successfully adapt to external changes (G. Klein & Pierce, 2001).

The team as a whole must be taught to engage in appropriate processes and collaboratively implement response repertoires. For many teams, one way to accomplish this objective is through simulated work environments. Simulators and simulations allow members to practice coordinating and experimenting with novel approaches without the risk of costly errors (Kozlowski, 1998; Senge, 1990). Following simulation practice, teams may be transferred to the actual work environment so that they may continue to hone their skills via action learning.

In summary, training team members and teams to be flexible, observant, and adaptive is an arduous and time-consuming undertaking for any organization. The complexities of training are only compounded when the emphasis is shifted toward developing entire teams that can adaptively coordinate to achieve shared goals. Nonetheless, adaptive teams can efficiently conserve resources, achieve synergistic process gains, and, in some contexts, save lives. To demonstrate how to fully reap these benefits, this article has described several techniques for enhancing training designed to facilitate adaptive team performance and team adaptation.

Directions for Future Research

One of the finest features of teams is that they have great potential for adaptation. With proper training and organizational support, teams may develop the capacity to surmount or prevent performance hurdles. In all likelihood, this would foster both team and organizational nimbleness (Conner & Hoopes, 1997). Because

of the present dearth of research investigating this topic, there are rich opportunities for further fine-tuning of the construct, the creation of measures, empirical testing, and ultimately the refinement of the theory advanced herein. It is hoped that the following discussion will stimulate interest for exploration in these areas.

Of primary importance to any future empirical investigations of team adaptation is the creation of adequate measures. Measuring any team-level variable presents a challenge, and creating a method for capturing instances of team adaptation would be no exception. Whereas it may be possible for trained subject matter experts to rate adaptive team behaviors during task performance, the less ostensible adaptations (e.g., changes in cognition and affect) must be measured by different methods, such as individual or team report. Obviously, these measures would have to be administered at multiple points in time in order to capture the unfolding process of change. Unfortunately, this method typically necessitates halting the team many times during the task to complete measures, thereby breaking members' concentration and possibly affecting their ensuing performance. This would be particularly deleterious for teams involved in stressful, resource-depleting tasks, as are often found in fluid operational environments. Thus, it is often difficult or impossible to observe cognitive and affective manifestations of adaptation.

As noted above, the main challenges to assessing team adaptation and the constructs that make up its nomological network include (a) assessing the covert (i.e., cognitive, affective) types of adaptation and (b) electing the best times to collect what often amounts to obtrusive measurements. Nevertheless, once these obstacles are overcome and a satisfactory measurement battery is developed, significant progress will be made toward comprehending and modeling the essence of team adaptation.

Fortunately, recent progress has been made in the area of automated speech recognition, which holds promise for circumventing many of the challenges associated with measuring the processes constituting adaptive team performance. Coupling automated speech recognition with latent semantic analysis provides unprecedented technological capability to unobtrusively and seamlessly model team performance (Foltz, Laham, & Derr, 2003). Such a system allows for real-time assessment and feedback, which would be particularly advantageous to both measuring and facilitating adaptive performance in dynamic contexts.

The model presented in Figure 1 displays the adaptive cycle within the input-throughput-output framework. To test the integrity of this model, it is necessary to conduct controlled laboratory experiments to explore the causality of the relationships among the primary variables that make up the adaptive cycle and their influence on team adaptation. Given that support is found for the advanced relationships, they must then be tested in the field to determine whether the effects found in the lab generalize to and across other settings, times, and persons (Shadish, Cook, & Campbell, 2002). It would also be fruitful to investigate whether the relationships among the primary variables hold across different types of teams (e.g., action teams, project teams).

Given that evidence is found for the predicted relationships among the primary variables, possible associations should be explored between the primary variables and secondary variables. For example, the relationship between task expertise and the situation assessment phase of the adaptive cycle could be examined with

additional analysis of the possible moderating effect of team type and interdependence level.

Conclusion

Teams are ubiquitous in public and private sector organizations. Understanding the nature of teams, teamwork, and team performance in order to promote team effectiveness has been an arduous endeavor undertaken in several scientific disciplines over the past century. In many contexts, team performance—and ultimately effectiveness—directly relates to how the team adapts to the multitude of contingencies that are encountered. For this reason, we have put forth a conceptual model of team-level adaptation, developed from a presently limited literature base. The advanced model is of theoretical importance in that it extends previous theoretical endeavors in at least three ways. First, the model defines the cyclical nature of adaptive team performance. In doing so, the model begins to answer a call for greater attention to time within team theory building and research by moving beyond a unidirectional aspect of time as typically depicted by input–process–output (IPO) models (e.g., Ilgen, Hollenbeck, Johnson, & Jundt, 2005; McGrath, 1964) to illustrate the recursive nature of both the adaptive cycle and the model as a whole. It is important to note that the speed of the episodic adaptive cycle will fluctuate, with increases in task intensity caused by natural, cyclical variations in temporal entrainments (e.g., Ancona & Chong, 1999; Kelly, Futoran, & McGrath, 1990; Kozlowski et al., 1999) as well as by contextual changes signaling the need for adaptation.

Second, the model outlines the inherent components and emergent nature of adaptive team performance. Several previous attempts to model aspects of team effectiveness have overemphasized IPO linkages while ignoring dynamic process interactions. In contrast, the advanced model provides insight into the internal workings of the process itself and highlights the centrality of emergent cognitive states resulting from, and serving as inputs to, the fluid activities undertaken by team members. Finally, the advanced model places this recursive process within a nomological net of antecedents and consequences within the familiar input–throughput–output framework. It is hoped that this model will serve as food for thought and initiate an expansion of the literature base on team adaptation, resulting in a broadened comprehension of the nature and practical applicability of team adaptation across various settings and team types.

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